UNIT-I

1. **A heat engine receives heat at the rate of 1500 kJ/min and gives an output of 8.2 kW. Determine the thermal efficiency and the rate of heat rejection.**

   Given:

   Heat received by engine (HS)

   Work output (W)

   Required: η_{th} and HR

   Solution:

   = 1500 kJ/min = 25 kW

   = 8.2 kW

   η_{th} = Work output / Heat Supplied = 8.2 / 25 = **0.328 ---- Ans**

   Workdone = HS – HR

   8.2 = 25 – HR

   Heat rejected (HR) = **16.8 kW**

2. **A heat engine receives heat at the rate of 1500 kJ/min and gives an output of 10.2 kW. Determine the thermal efficiency and the rate of heat rejection.**

   Given:

   Heat received by engine (HS)

   Work output (W)

   Required: η_{th} and HR

   Solution:

   = 1500 kJ/min = 25 kW

   = 10.2 kW

   η_{th} = Work output / Heat Supplied = 8.2 / 25 = **0.328 ---- Ans**

   Workdone = HS – HR

   8.2 = 25 – HR

   Heat rejected (HR) = **20.8 kW ---- Ans**

3. **0.04 kg of CO (M = 44) is compressed from 1 bar, 20°C, until the pressure is 9 bar and the volume is then 0.003 m^3. Calculate the change of entropy. Take C_p for CO_2 as 0.88 kJ/K and assume CO_2 to be a perfect gas.**

Mass of gas (m)

Initial pressure (p_1)

= 0.04 kg

= 1.05 bar

Initial temperature (T_1) = 20°C = 293 K

Final pressure (p_2) Final volume (V_2) *Required*: ⊗ S

Solution:

= 9 bar

= 0.003 m^3

$$S_2 - S_1 = m\,R \ln [V_2/V_1] + m\,C_v \ln [T_2/T_1]$$ ⇗ General equation

Universal gas constant (R_u) = 8314 J/kg-K

Characteristic gas constant (R) = R_u/M = 8314/44 = 188.95 J/kg-K

To find m

$$p_1 V_1 = m R T_1$$

$$1 \times 10^5 \times V_1 = 0.04 \times 188.95 \times 293$$

$$V_1 = 0.02214494\ m^3$$

To find T_2

$$p_2 V_2 = m R T_2$$

$$9 \times 10^5 \times V_2 = 0.04 \times 188.95 \times T_2$$

$$T_2 = 357.24\ K$$

To find C_v

$$C_p - C_v = R$$

$$880 - C_v = 188.95$$

$$C_v = 691.05\ J/kg\text{-}K$$

$$S_2 - S_1 = (0.04)\,(188.95)\ln[0.003/0.02214494] + (0.04)\,(691.05)\ln[357.24/293]$$

= **-9.62847 J/kg-K --- Ans**

UNIT-2

1. **In a Carnot cycle, the maximum pressure and temperature are limited to 18 bar and 410°C. The ratio of isentropic compression is 6 and isothermal expansion is 1.5. Assuming the volume of the air at the beginning of isothermal expansion as 0.18 m^3, determine (i) the pressure and temperature at main points (ii) change in entropy during isothermal expansion (iii) mean thermal efficiency of the cycle (iv) mean effective pressure of the cycle and (v) the theoretical power if there are 210 working cycles per min.**

   Given:

   Maximum pressure (p_3) = 18 bar

   Maximum temperature (T_3) = 410°C = 683 K = T_4

   Ratio of isentropic compression (V_2/V_3) = 6

(ii) Change in entropy during isothermal expansion

$$\otimes S = m R \ln [V_4/V_3]$$

To find m

$$p_1 V_1 = m R T_1$$

To find V_1

$$V_1 = 6 \times 0.27 = 1.62 \text{ m}^3$$

$$0.977 \times 10^5 \times 1.62 = m \times 287 \times 333.55$$

(iii)

(iv) $p_m = W / V_s$

m = 1.653 kg

S = 1.653 x 287 x ln [1.5] = **192.36 J/K ---- Ans** | = (T_{max} - T_{min}) / T_{max} = (683 – 333.55) / 683

= **0.5116 ---- Ans**

W = Area of T-s chart = $\otimes$ S ($T_{max} - T_{min}$) = 192.36 x (683 – 333.55)

= 67220.2 J $V_s = V_1 - V_3$

= 1.62 – 0.18 = 1.44 m^3

p_m = 67220.2 / 1.44 = **46680.7 bar --- Ans**

Power = W x n / 60 = 67220.2 x 210 / 60 = **2357270.7 W --- Ans**

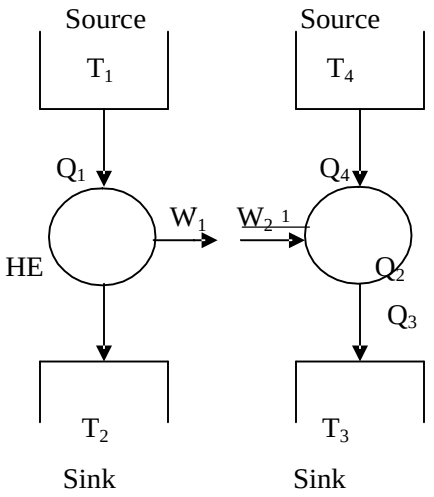
2. A reversible heat engine operates between two reservoirs at temperatures of 600°C and 40°C. The engine drives a reversible refrigerator which operates between reservoirs at temperatures of 40°C and -20°C. The heat transfer to the engine is 2000 kJ and the net work output of the combined engine-refrigerator plant is 360 kJ. (a) Evaluate the heat transfer to the refrigerant and the net heat transfer to the reservoir at 40°C. (b) Reconsider (a) given that the efficiency of the heat engine and the COP of the refrigerator are each 40% of their maximum possible values.

Given:

Combined engine-refrigerator plant

Engine source temperature (T_1)	= 600°C = 873 K
Engine sink temperature (T_2)	= 40°C = 313 K
Heat transfer to the engine (Q_1)	= 2000 kJ
Refrigerator source temperature (T_4)	= -20°C = 253 K
Refrigerator sink temperature (T_3)	= 40°C = 313 K
Net work transfer ($W_1 - W_2$)	= 360 kJ

Required: (a) For reversible engine, Heat transfer to the refrigerant (Q_4) and Net heat transfer to the reservoir at 40^oC ($Q_2 + Q_3$)



$$\eta_{rev} = \frac{873 - 313}{873} = 0.0414$$

$$0.6414 = \frac{W}{2000} \qquad W_1 = 1282.8 \text{ kJ}$$

$W_1 = Q_1 - Q_2$

$1282.8 = 2000 - Q_2 \qquad Q_2 = 717.2$ kJ

$W_1 - W_2 = 360$

$1282.8 - W_2 = 360 \qquad W_2 = 922.8$ kJ

$$COP_{rev} = \frac{Q_4}{Q_3 - Q_4} = \frac{Q_4}{W_2} = \frac{T_4}{T_3 - T_4}$$

$$COP_{rev} = \frac{253}{313 - 253} = 4.2166$$

$$4.2166 = \frac{Q_4}{922.8} \qquad Q_4 = \textbf{3891 kJ} \quad \text{----- } \textbf{Ans}$$

$W_2 = Q_3 - Q_4$

$922.8 = Q_3 - 3891 \qquad Q_3 = 4813.8$ kJ

Therefore, $Q_3 + Q_4 = 4813.8 + 717.2 =$ **5531 kJ ------ Ans**

(b)

$$\eta = 0.4\eta_{rev} = 0.4(0.6414) \quad 0.256$$

$$COP = 0.4COP_{rev} = 0.4(4.2166) \quad 1.686$$

$$W$$

$$0.256 = \frac{}{2000} \quad W_1 = Q_1 - Q_2$$

$$512 = 2000 - Q_2 \quad W_1 - W_2 = 360$$

$$512 - W_2 = 360$$

$$1.686 = \frac{Q_4}{152} \quad W_2 = Q_3 - Q_4$$

$$152 = Q_3 - 256.3 W_1 = 512 \text{ kJ}$$

$Q_2 = 1488$ kJ

$W_2 = 152$ kJ

$Q_4 =$ **256.3 kJ ----- Ans**

$Q_3 = 408.3$ kJ

Therefore, $Q_3 + Q_4 = 1488 + 408.3 =$ **1896.3 kJ ------ Ans**

UNIT-3

A heat engine operating between two reservoirs at 1000 K and 300 K is used to drive a heat pump which extracts heat from the reservoir at 300 K at a rate twice that at which the engine rejects heat to it. If the efficiency of the engine is 40% of the maximum possible and COP of the heat pump is 50% of the maximum possible, what is the temperature of the reservoir to which the heat pump rejects heat? What is the rate of heat rejection from the heat pump if the rate of heat supply to the engine is 50 kW?

Given:

Combined engine-heat pump plant

Engine source temperature (T_1) = 1000 K

Engine sink temperature (T_2) = 300 K

Heat supply to the engine (Q_1) = 50 kW

Heat pump source temperature (T_4) = 300 K

Efficiency of the engine (η) = 0.4 η_{rev}

COP of the heat pump $\frac{1}{}$ = 0.5 COP_{rev}

Hear extraction by the heat pump (Q_4) = 2 x Heat rejection by the engine (Q_2)

Required: Temperature of heat pump sink (T_3) and Heat rejection from the pump (Q_3)

Solution:

(a)

$$| \ _{rev} = \frac{T_1 \quad T_2}{T} = \frac{W_1}{Q} = \frac{Q_1 \quad Q_2}{Q}$$

$$0.28 = \frac{W_1}{50}$$

$W_1 = Q_1 - Q_2$

$14 = 50 - Q_2$

$W_1 = 14$ kW

$Q_2 = 36$ kW

$Q_4 = 2\ (36) = 72$ kW

$$COP_{rev} = \frac{Q_3}{Q_3 \quad Q_4} = \frac{Q_3}{W_2} = \frac{T_3}{T_3 \quad T_4}$$

$W_2 = Q_3 - Q_4$

$14 = Q_3 - 72$ $Q_3 =$ **86 kW ---- Ans**

Note: $W_1 = W_2$

$$COP = \frac{Q_3}{Q_3 \quad Q_4} = \frac{Q_3}{W_2}$$

$$COP = \frac{86}{14} = 6.1428$$

$$COP_{rev} = \frac{COP}{0.5} = \frac{6.1428}{0.5} = 12.286$$

$$12.286 = \frac{T_3}{T_3 \quad 300}$$ $T_3 =$ **326.6 K ----- Ans**

. Air expands through a turbine from 500 kPa, 520°C to 100 kPa, 300°C. During expansion 10 kJ/kg of heat is lost to the surroundings which is at 98 kPa, 20°C. Neglecting the KE and PE changes, determine per kg of air (a) the decrease in availability, (b) the maximum work, and (c) the irreversibility.

Given:

Open system

Initial pressure of air (p_1) = 500 kPa = 500 x 10^3 Pa

Initial temperature of air (T_1) = 520°C = 793 K

Final pressure of air (p_2) Final temperature of air (T_2) Environment pressure (p_o)

Environment temperature (T_o)

= 100 kPa = 100 x 10^3 Pa = 300°C = 573 K

= 98 kPa = 98 x 10^3 Pa

= 293 K

Heat loss to the surroundings (q) = 10 kJ/kg = 10 x 10^3 J/kg

Required: (a) $\Phi_1 - \Phi_2$ (b) W_{max} (c) I

Solution:

(a) $\Phi_1 - \Phi_2 = (H_1 - H_2) - T_o(S_1 - S_2)$

$= (h - {}_1h_2) - T_o(s_1 - s_2)$ per kg

$h - {}_1h_2 = C_p(T - {}_1T_2) = (1005)(793 - 573) = 221100\ J/kg$

$s - {}_1s_2 = R\ln\left(\frac{V_1}{V_2}\right) + C_v \ln\left(\frac{T_1}{T_2}\right)$

$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$

$\frac{V_1}{V_2} = \frac{p_2 T_1}{p_1 T_2} = \frac{100 x 793}{500 x 573} = 0.2768$

$s_1 - s_2 = (287)\ln(0.2768) + (718)\ln\left(\frac{793}{573}\right) = 135{,}34\ J/kgK$

$= (221100) - 293(-135.34) = 260754.62\ J/kg$ --- **Ans**

(b) $w_{max} = \Phi_1 - \Phi_2 = 260754.52\ J/kg$ ---- **Ans**

(c) Irreversibility, I = w_{max} - w_{act}

SFEE equation is given by,

m h_1 + m q = m h_2 + m w_{act}

w_{act} = q + (h_1 – h_2) = (-10 x 10^3) + (221100)

= 211100 J/kg

I = 260754.62 – 211100 = **49654.62 J /kg** ---- **Ans**

Air Standard Cycles

Air Standard cycles: Cycles using a perfect gas, having the properties of air useful in the study of the I. C. Engine because they represent a limit to which actual cycle may approach and they are subjected to simple mathematical and explanatory treatment.

Assumptions made for analysis:

- The properties of the working medium can be calculated by the application of the perfect gas equation.

  i.e., $pV = mRT$

- The Specific heat of the substance remains same during all the processes in the cycle.

  i.e.,C_p & C_v are unchanged.

- The cycles are composed of reversible processes.

- The gas does not undergo any chemical changes.

Various cycles

1. Carnot cycle
2. Otto cycle
3. Diesel cycle
4. Stirling cycle
5. Brayton cycle
6. Erricson cycle
7. Dual cycle

Air standard efficiency of a cycle

The thermal efficiency of an ideal air standard cycle is called the "Air standard efficiency".

In an ideal air standard cycle, the working fluid is air. The petrol and diesel engines working on Otto cycle and diesel cycle use petrol and diesel oil with air. This air fuel mixture behaves like air before the combustion takes place. The properties of combustion products are also not different from those of air. Therefore the efficiencies of petrol and diesel engines are calculated assuming them working on air standard cycles.

The efficiency of a cycle is given by,

$$\eta = \frac{\text{Output}}{\text{Input}} = \frac{\text{Net work output}}{\text{Heat Supplied}}$$

$$= \frac{\text{Heat supplied} - \text{Heat rejected}}{\text{Heat supplied}}$$

Carnot Cycle:

The Carnot cycle is a particular thermodynamic cycle, modeled on the Carnot heat engine, studied by Nicolas Léonard Sadi Carnot in the 1820s and expanded upon by Benoit Paul Émile Clapeyron in the 1830s and 40s.

This cycle has the highest possible efficiency and consists of four simple operations:

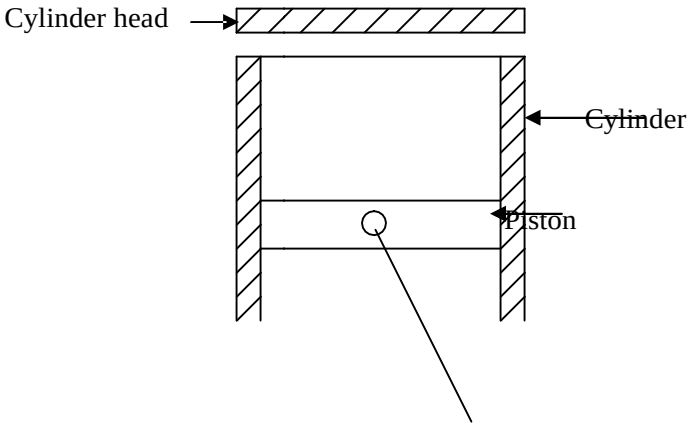
(i) Isothermal expansion

(ii) Isentropic expansion

(iii) Isothermal compression

(iv) Isentropic compression

Operation: Consider a cylinder piston arrangement as shown in the fig. Let m kg of air is enclosed in a cylinder. The cylinder head is made of perfect heat conductor or perfect heat insulator. The cylinder is perfectly insulated.



The cylinder is full of air when the piston is at BDC. The 'perfect heat conductor cylinder head' (Cold body) is brought in contact with the cylinder. The air is compressed at constant temperature from (1) to (2) during its travel towards TDC. During this process the heat is rejected from the air. The 'cold body' is removed and 'perfect heat insulator head' is brought in contact with the cylinder. Now the air is compressed isentropically (2-3) till the piston reaches the TDC. Then the perfect insulator is removed. The perfect conductor (Hot body) is brought in contact with the cylinder and the heat is supplied to the air. Now expansion proceeds at constant temperature (3-4) during its travel towards BDC. The perfect insulator is brought and further expansion proceeds isentropically upto BDC (4-1). Thus the cycle is completed.

Efficiency of cycle

Process – 1 – 2 ⇗ Isothermal compression

⇗ Heat transferred = $Q_{1\text{-}2} = m\ T_2\ (s_2 - s_1) = HR$

Note: The area under the curve in T-s diagram is heat transferred.

Process – 2 – 3 ⇗ Isentropic compression

- Heat transferred = $Q_{2-3} = 0$;

Process – 3 – 4 ⇗ Isothermal expansion

- Heat transferred = $Q_{3-4} = m\ T_1\ (s_2 - s_1) = HS$

Process – 4 – 1 ⇗ Isentropic expansion

- Heat transferred = $Q_{4-1} = 0$

- Cycle efficiency ( ) = [Heat supplied – Heat rejected]/Heat supplied

$$= \frac{m\ T_1\ (s_2 - s_1) - m\ T_2\ (s_2 - s_1)}{m\ T_1\ (s_2 - s_1)}$$

$$= \frac{T_1 - T_2}{T_1} = \frac{T_{max} - T_{min}}{T_{min}}$$

The Carnot cycle efficiency is depending only on the temperatures T_1 and T_2.

Reversed Carnot cycle

p-V and T-s diagrams of reversed Carnot cycle are shown. The cylinder is full of air at temperature T_2 when the piston is at BDC. The 'perfect heat insulator cylinder head' is brought in contact with the cylinder. The air is compressed isentropically from (1) to (2) during its travel towards TDC. During this process the temperature of air is raised to T_1. The 'perfect heat insulator is removed and 'perfect heat conductor head' (Cold body) is brought in contact with the cylinder. Now the air is compressed at constant temperature (2-3) till the piston reaches the TDC. Then the cold body is removed. The perfect insulator is brought in contact with the cylinder. Now the air expanded isentropically from (3) to (4) during its travel towards BDC. During this process the temperature of air is lowered to T_2. The perfect conductor (Hot body) is brought and further expansion proceeds at constant temperature upto BDC (4-1). Thus the cycle is completed.

Efficiency of cycle

Process – 1 – 2 ⇗ Isentropic compression

- Heat transferred = $Q_{1-2} = 0$

Note: The area under the curve in T-s diagram is heat transferred.

Process – 2 – 3 ⇗ Isothermal compression

- Heat transferred = $Q_{2-3} = m\ T_1\ (s_2 - s_3) = HR$;

Process – 3 – 4 ⇗ Isentropic expansion

- Heat transferred = $Q_{3-4} = 0$;

Process – 4 – 1 ⇗ Isothermal expansion

- Heat transferred = $Q_{4-1} = m\ T_2\ (s_1 - s_4) = m\ T_2\ (s_2 - s_3) = HS$;

4 COP = Refrigeration effect / Work input

= Heat supplied to the air / (Heat rejected – Heat supplied)

$$= \frac{m\, T_2 (s_2 - s_2)}{m\, T_1 (s_2 - s_3) - m\, T_2 (s_2 - s_3)}$$

$$= \frac{T_2}{T_1 - T_2} = \frac{T_{min}}{T_{max} - T_{min}}$$

The Carnot cycle COP is depending only on the temperatures T_1 and T_2.

Absolute Thermodynamic temperature scale (Kelvin scale)

The efficiency of any heat engine receiving heat Q_1 and rejecting heat Q_2 is given by

$$\eta = \frac{W_{net}}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

The efficiency of any reversible heat engine is given by

$$\eta_{rev} = \frac{W_{net}}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

$$\frac{Q_2}{Q_1} = f(T_1, T_2)$$

Or in terms of new function F, $\frac{Q_1}{Q_2} = F(T_1, T_2)$

Consider reversible heat engines operating in series.

We can write, $\frac{Q_1}{Q_2} = F(T_1, T_2)$

$$\frac{Q_2}{Q_3} = F(T_2, T_3)$$

$$\frac{Q_1}{Q_2} = \frac{Q_1 / Q_3}{Q_2 / Q_3} = \frac{F(T_1, T_3)}{F(T_2, T_3)} = \frac{(T_1)}{(T_2)}$$

$$Q = T$$

1compression refrigeration system uses R-12 and operation pressure limits of 0.745 and 0.15 MPa. The vapour entering the compressor has a temperature of –10°C and the liquid leaving the condenser is at 28°C. A refrigerating load of 2 kW is required. Determine the COP and the swept volume of the compressor if it has a volumetric efficiency of 76% and runs at 600 rpm.

Heat Engine, Heat Pump

Heat engines, Refrigerators, Heat pumps:

- A heat engine may be defined as a device that operates in a thermodynamic cycle and does a certain amount of net positive work through the transfer of heat from a **high temperature body to a low temperature body.** A steam power plant is an example of a heat engine.
- A refrigerator may be defined as a device that operates in a thermodynamic cycle and transfers a certain amount of heat from a body at a **lower temperature to a body at a higher temperature** by consuming certain amount of external work. Domestic refrigerators and room air conditioners are the examples. In a refrigerator, the required output is the heat extracted from the low temperature body.
- A heat pump is similar to a refrigerator, however, here the required output is the heat rejected to the high temperature body.

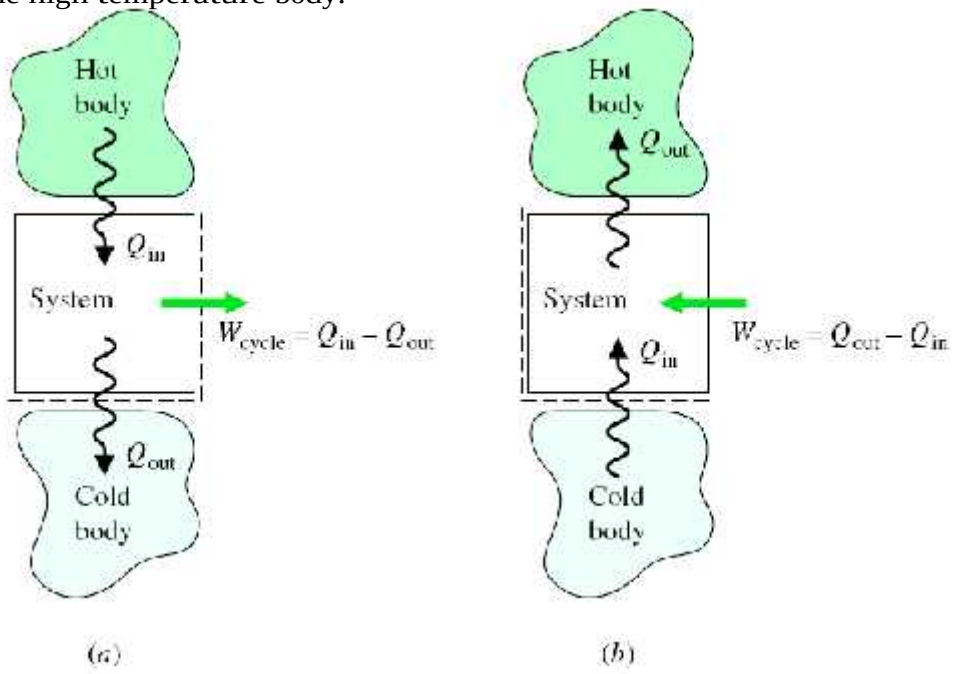


Fig. (*a*) Heat Engine (*b*) Refrigeration and heat pump cycles

2.A 100 tonne low temperature R-12 system is to operate on a 2-stage vapour compression refrigeration cycle with a flash chamber, with the refrigerant evaporating at – 40°C, an intermediate pressure of 2.1912 bar, and condensation at 30°C. Saturated vapour enters both the compressors and saturated liquid enters each expansion valve. Consider both stages of compression to be isentropic. Determine:

(a) The flow rate of refrigerant handled by each compressor (b) The total power required to drive the compressor

(c) The piston displacement of each compressor, if the clearance is 2.5% for each machine

(d) The COP of the system

(e) What would have been the refrigerant flow rate, the total work of compression, the piston displacement in each compressor and the compressor and the COP, if the compression had occurred in a single stage? .

(**Ans.** (a) 2.464, 3.387 kg/s, (b) 123 kW, (c) 0.6274, 0.314 m^3/s, (d) 2.86, (e) 3.349 kg/s, 144.54 kW, 1.0236 m^3/s, 2.433)

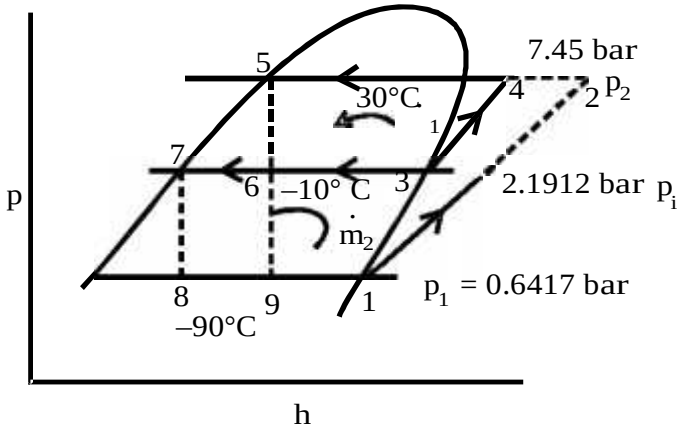
Solution:

$h_1 = 169$ kJ/kg

$h_3 = 183.2$ kJ/kg

$h_5 = 64.6$ kJ/kg $= h_6$

$h_7 = h_8 = 26.9$ kJ/kg



$S_1 = S_2 = 0.7274$ kJ/kg – K

$S_3 = S_4 = 0.7020$ kJ/kg – K

4 From P.H chart of R12

$h_2 = 190$ kJ/kg

$h_4 = 206$ kJ/kg

$$\dot{m}_2 h_2 + \dot{m}_1 h_5 = \dot{m}_2 h_7 + \dot{m}_1 h_3$$

4 $$\dot{m}_2(h_2 \quad h_7) = \dot{m}_2(h_3 \quad h_5)$$

4 $$\frac{\dot{m}_1}{\dot{m}_2} \quad \begin{matrix} h_2 & h_7 \\ h_3 & h_5 \end{matrix} = \frac{190 \quad 26.9}{183.2 \quad 64.6} = 1.3752$$

$m_2 = \frac{67 \text{ kg/s}}{3600}$

$m_1 = m_2 \times 1.3752 = 3.7635$ kg/s

(b) Power of compressor (P) = $m_2(h_2 - h_1) + m_1(h_4 - h_3)$ = 14328 kW

(d) COP = $\frac{\text{Refrigeration efficiency}}{\text{Compressor}} = \frac{100 - 14000}{3600 - 1}$ = 2.7142

(e) For single storage

From R12 chart h_{a2} = 2154 kJ/kg, h_g = h_s = 64.6 kJ/kg

$m(h_1 - h_9) = \frac{100 - 14000}{3600}$

$\therefore m = 3.725$ kg/s

Compressor power (P) = $m(h_{42} - h_1)$ = 3.725 × 46 = 171.35 kW

UNIT-5

A Refrigerant-12 vapour compression cycle has a refrigeration load of 3 tonnes. The evaporator and condenser temperatures are – 20°C and 40°C respectively. Find

(a) The refrigerant flow rate in kg/s

(b) The volume flow rate handled by the compressor in m³/s

(c) The work input to the compressor in kW

(d) The heat rejected in the condenser in kW

(e) The isentropic discharge temperature.

If there is 5° C of superheating of vapour before it enters the compressor, and 5° C sub cooling of liquid before it flows through the expansion valve, determine the above quantities.

As 50°C temperature difference in evaporate so evaporate temperature = – 20°C and

Condenser temperature is 30°C.

p_1 = 1.589 bar

p_2 = 7.450 bar

h_7 = 178.7 kJ/kg, h_3 = 64.6 kJ/kg

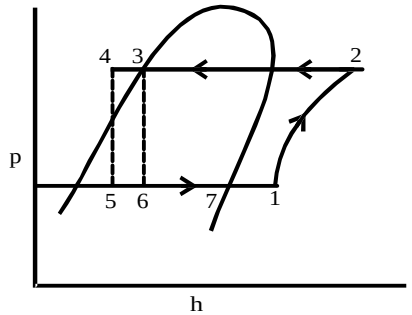
$h_1 = 178.7 + \frac{5}{20}(190.8 - 178.7)$

$\otimes h$ = 3.025 kJ/kg

$s_1 = 0.7088 + \frac{5}{20}(0.7546 - 0.7088)$

= 0.7203 kJ/kg– K

$h_3 - h_4 = \otimes h = h_1 - h_7 = 3.025$



$h_4 = h_3 - \otimes h$ = 61.6 kJ/kg i.e.

4 Degree of sub cooling = 3.06°C [Data from CP Arora]

25°C $h_g = 59.7$
30°C $h_g = 64.6 \square 0.98/v_c$

(a) Degree of super heat is discharge = $\frac{0.7203 - 0.6854}{0.7321 - 0.6854} \times 20$ =

15°C ∴ Discharge temperature = 15 + 30 = 45° C

∴ $h_2 = 199.6 + \frac{15}{20}(214.3 - 199.6) = 210.63$ kJ/kg

∴ Compressor work (W) = $h_2 - h_1$ = 210.63 – 181.73 = 28.9 kJ/kg

Refrigerating effect (Q_0) = $h_7 - h_5 = h_7 - h4$ = (178.7 – 61.6) kJ/kg = 117.1 kJ/kg